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CIVIL & ENVIRONMENTAL ENGINEERING | RESEARCH ARTICLE

Economic viability and environmental sustainability of a grid-connected solar PV plant in Yaounde - Cameroon using RETScreen expert

Chu Donatus Iweh^{1*}, Samuel Gyamfi¹, Emmanuel Tanyi² and Eric Effah-Donyina¹

Abstract: The Cameroon power sector is currently undergoing a period of transition with government setting ambitions to increase the generation of clean electricity to meet the rapidly growing demand. A significant aspect of this transition is the phasing out of some thermal plants and supporting power generation with renewables. This can be achieved through the heavy exploitation of the renewable energy resources in the country. In line with this goal, the study assesses the feasibility of a 211.75 MW solar PV power plant in Yaounde, Cameroon using RETScreen Expert. The simulation showed an annual electricity production of 304,668.191 MWh with arrays mounted on a fixed axis. The model considered a debt ratio of 40%, debt interest rate of 12%, PV cost per kW of \$1371 and a total system initial cost of \$291,791,500. The annual revenue from exporting power to the grid was \$36,560,183 and a capacity factor of 16.4%. The solar PV project was economically viable with a cost of energy (COE) of \$75.43/MWh or \$0.075/kWh and a gross annual GHG emission reduction potential of 61,004.5 tCO₂ corresponding to 141,870.9 barrels of crude oil not used throughout the project life time. The benefit-cost ratio obtained in the simulation was 4.5 which implies that the project is profitable (ratio is greater than 1). The project break-even point of equity payback was 9.2 years while the simple payback was 8.9 years. The PV system had an IRR-assets of 7.4%, which was below the debt interest rate of Cameroonian financial institutions hence, the project will be attractive to investors. Also, the project risk and sensitivity analysis showed that a reduction in the project's initial cost had a significant effect on the net present value (NPV) and the cost of energy production. Similarly, a rise in the grid electricity export reduced the cost of energy production when the confidence level was set to 90%.

Subjects: Environmental Sciences; Environmental Studies; Environmental Issues; Environment & Resources; Environment & Economics; Engineering Economics; Energy policy and economics; Solar energy; Sustainability Assessment; Environmental Economics

Keywords: RETScreen; Sensitivity; electricity export; GHG emissions; break-even; Net Present Value; IRR

1. Introduction

Cameroon is currently suffering from severe power shortage which has greatly slowed down economic growth and has increased hardship. Power production in Cameroon is mostly from locally-sourced resources, with hydropower dominating the generation mix (63%) while oil and gas complement the electricity demand. There are huge potentials for power export to

neighboring countries due to the abundance of renewable energy (RE) resources. The over reliance of Cameroon on hydropower has further exacerbated the power crisis especially in the dry season where droughts will often cause prolonged power outages. This situation was glaring in 2015 when an outage disrupted work at the Douala port (Kindzeka, 2022) for a couple of days inflicting huge financial losses to users who could not recover their merchandise on time (Kindzeka, 2019). Another incident was that which occurred during the just ended 2022 African Nations Cup, hosted by the country, where industries were asked to reduce production as more power was reserved for football stadia and residential consumers. Government's intervention to avoid future scenarios has been the ambition to diversify the power mix with an increase in generation through a number of economic incentives for renewables. Experts argue that the power unreliability concerns in Cameroon (World Bank Group (WBG), 2017) are mostly caused by the generation mix and grid infrastructure than the management of the utility. Another challenge is the deterioration of the utility's financial and technical status. The utility operator suffers from increasing cost of fossil fuel procurement for thermal plants as well as high technical losses caused by the aging grid infrastructure.

Recently, Cameroon is advancing economic incentives to support the growth of solar PV such as a 10-year tax break on solar PV projects (Ngalame, 2022) and the waiving of the value added tax (VAT) on imported solar accessories (Presidency of the Republic of Cameroon (PRC), 2011). Despite the measures put in place to encourage private investment, there is a high perceived risk in the sector and this is one of the reasons why there is currently no grid-connected solar PV system in Cameroon (Iweh et al., 2022). Even with the high solar potentials (Institute of Electrical and Electronic Engineers (IEEE), 2018) owing to Cameroon's location, there is an insignificant capacity of 14.19 MW installed in off-grid sites (International Renewable Energy Agency (IRENA), 2019a). Literature on the financial viability of grid connected solar PV systems in Cameroon is scanty and little is known by investors on whether solar PV development could be a profitable business in Cameroon. The new economic incentives presented by the government are attractive but do not seem to be creating the much-anticipated impact. Hence, there is need for more studies that assesses the economic viability of solar PV systems in Cameroon so as to keep stakeholders informed on the business opportunity. Exploiting solar energy for electricity generation is a sustainable option in Cameroon since it is a non-polluting renewable energy resource. Moreover, Cameroon's geographic position around the equator exposes her to a significant amount of solar radiation and hence the implementation of cost-effective solar PV systems would lower the electricity crisis as well as reduce the cost of power generation in the country. Among the various renewable energy sources in the country, solar PV is one of the fast-growing energy sources. Therefore, it is imperative to invest in the highly under-exploited solar resource for sustainable development. This will reduce the power deficit as well as lessen the use of peaking thermal plants. The government is faced with financial constraint which inhibits her investment in power expansion projects. However, there are possibilities to attract international investors and donors through developing an investment-friendly solar PV road map consisting of an inclusive feasibility analysis exploring the financial risk, technical, sensitivity and environmental effect of solar PV in Cameroon. A clean energy management tool that can effectively conduct this analysis is the RETScreen Expert software. This tool is the property of the Canadian government used to conduct techno-economic and environmental studies in energy systems. This paper examines the feasibility of deploying a grid-connected solar PV in Yaounde, Cameroon so that the results could be used to persuade solar PV investors to consider investing in solar PV projects in Cameroon. A solar PV capacity of 211.75 MW is considered for this study. This is only an example because the solar irradiance in the city is capable of generating more electricity with a high Return on Investment as well as a high greenhouse gas (GHG) emission reduction potential. This will help support Cameroon's commitment to the Paris Agreement as indicated in her Nationally Determined Contributions (United Nations Framework Convention on Climate Change (UNFCCC), 2022).

The paper starts with the presentation of Cameroon's electricity challenge where the aspect of over dependence on hydropower and the problem of low solar PV uptake is highlighted. Previous

studies on the use of RETScreen in solar PV development are discussed in order to establish the contribution of the study. This was followed by a brief discussion on the solar potentials of Cameroon. Furthermore, the methodology of the study was presented. This section involved the discussion of the RETScreen model and the site characteristics. Finally, the results of the simulation were presented where the 211.75 MW grid-connected solar PV showed positive NPV and relatively shorter payback period of 8.9 years. The study ended with a conclusion, summarizing the major findings.

1.1. Literature on the use of RETScreen in solar PV deployment

Several scholars have conducted analysis on solar PV with the use of tools such as RETScreen, PVSyst, HYBRID, and HOMER. These studies have assessed the economic viability of deploying both off/on grid electrification systems. An assessment of the economic and environmental feasibility of a solar PV plant was conducted by (Rashwan et al., 2017) with the aim of transforming the grid into 100% solar PV power system to be used for residential buildings in Saudi Arabia. The RETScreen tool was used to evaluate the PV plant while comparing the government electricity tariff and the commercial sector of 4.0 and 8.0 cents/kWh respectively, with that of the international PV project model. Sensitivity and risk assessment was not considered in the study. Similar research was done by (Rehman et al., 2017) using the RETScreen tool where the feasibility of implementing a 10 MW grid-connected solar PV plant in Saudi Arabia was conducted. The data used in the model included relative humidity, global solar radiation, sunshine duration and dry bulb temperature. The output of the model included parameters such as greenhouse gas (GHG) emission reduction potential, energy produced, payback period, and net present value (NPV). The simulated model helped in the selection of a single solar PV system. Similarly, the scholars in (Mehmood et al., 2014) used the RETScreen software in the modeling of the load demand of Pakistan's residential sector powered through solar PV technology. They conducted a comparative analysis among parameters such as solar irradiance, load correlation, and solar fraction under ambient conditions. The model was evaluated based on indicators such as GHG emissions reduction potential, NPV, Internal Rate of Return (IRR) and payback period. They concluded that it was viable to implement a 5 kW off-grid solar PV in the area under usual climatic conditions with the possibility of reducing 0.6–0.7tCO₂ GHG emissions. However, this study was a standalone system aimed at meeting a predefined load and it was void of the challenges involved in grid interconnection. While this study considered the environmental and economic analysis, it failed to include sensitivity and risk assessment. A study to design renewable energy (RE) systems was conducted in (Thevenard et al., 2000) where modeling procedures were presented. Aspects considered in the design included number of solar arrays and battery. The system parameters were continuously varied in order to determine the most effective technology depending on load, climatic conditions and season. Similarly, the study was focused on meeting the power demand in an economically viable manner and failed to conduct analysis on the risk factors and sensitivity of the project. Technical, financial and environmental assessment of solar PV plants were carried out in Iran with the RETScreen software by (Mirzahosseini & Taheri, 2012) where the scholars proposed a new electricity tariff on the basis of three scenarios. Sensitivity and risk analysis as well as the GHG emission reduction credit scheme were not included in their analysis. The authors in (Khandelwal & Shrivastava, 2018) used the RETScreen tool to assess the environmental and financial viability of a solar PV plant for a rural community in India. The load supplied by the solar PV plant was 587.7 kW. This was a standalone system focused on meeting the load of a rural community with no considerations on how grid interconnection could affect the economic parameters in the project.

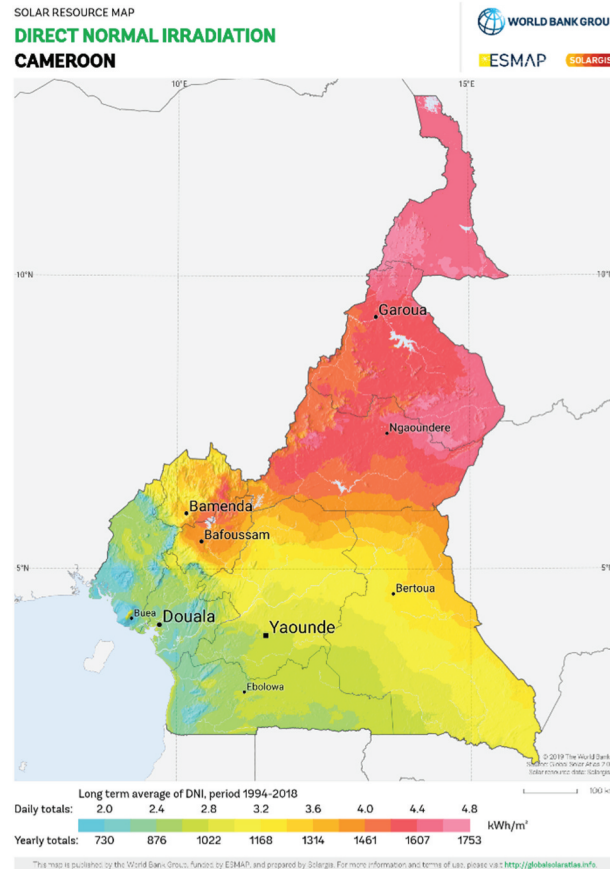
From the review, it can be seen that economic, technical and environmental assessment of energy systems in Cameroon using RETScreen are scarce. The only known study is the assessment of wind and solar resources using RETScreen (Tansi, 2011). The study focused on the technical feasibility as well as the environmental and economic viability of the solar and wind power systems. Most of the studies in Cameroon have worked on the economic viability of off-grid hybrid power system including solar PV (Nfah & Ngundam, 2009, (Nfah et al., 2008), Kenfack et al., 2009), mostly using HOMER or other economic assessment-based tool. This shows that less studies on the

analysis of the sustainability of a grid-connected solar PV system have been conducted in Cameroon using RETScreen. This study investigates the economic viability of solar PV systems in Cameroon as a means to advance the deployment of solar PV in the country. Most studies have either been based on the assessment of renewable energy (RE) potential (Ayompe & Duffy, 2014; Kaoga et al., 2016; Kewir et al., 2021; Nemzoue et al., 2020; Njom, 2016) or the utilization of off grid renewables to solve the challenge of rural electrification (Iweh & Lemundem, 2019; Nfah & Ngundam, 2009; Njoh et al., 2019). This study will evaluate the impact of some financial and technical parameters on capital cost, simple payback period, NPV and COE in the Cameroonian context in order to inform investors on the available potentials. There is need for suitable economic, risk and sensitivity assessment to erase the flawed perception that solar power investment in Cameroon is risky and expensive. Another observable gap in literature is that so much attention is given to the economic, environmental and the technical feasibility of solar PV using RETScreen software by scholars, with no consideration on the GHG emission reduction credit scheme, sensitivity and the risk analysis. These are essential aspects in project evaluation where the uncertainty of some project parameters needs to be analyzed to predict how some variables affect the sustainability of the project. However, few scholars attempted to include the aspect of sensitivity and risk assessment in their studies such as (Bustos et al., 2016) in the analysis of a solar PV plant in Chile. A total of 5 parameters and a variation of $\pm 30\%$ were considered in the study and the results revealed that the electricity export rate and the initial cost greatly affected the project outcome. Similarly, the scholars in (Harder & Gibson, 2011) used the same approach to assess the cost and revenue of a solar PV plant in Abu Dhabi. Findings from their analysis showed that the low expected price of electricity produced and the high initial cost hindered the implementation of solar PV systems in Abu Dhabi. Also, sensitivity and risk assessment were used by (Alam & Sadrul, 2011) to evaluate a grid-connected solar PV resource potential and feasibility in Bangladesh. The sensitivity analysis showed that the net present value (NPV) varied considerably with changes in the electricity export rate and the project initial cost. Another recent study that considered sensitivity and risk assessment was conducted in Nigeria by (Owolabi et al., 2019) using the RETScreen tool. They assessed the viability of a 6 MW grid-connected solar PV system. Six sites were considered in the analysis while other aspects such as the technical, economic, and environmental issues of the project were also included in their analysis. It is seen that the scholars that considered sensitivity and risk assessment in their studies did not include the possible implementation of a GHG emission reduction credit scheme that could encourage people to actively participate in renewable energy development. It is hardly discussed, especially in developing countries located in Sub Sahara Africa, and it has been considered in this study in order to ignite policy debates in Cameroon on the adoption of this potentially promising scheme. This study constitutes a singular attempt to examine the feasibility of developing large scale grid-connected solar PV systems in Cameroon. It proffers informed perspectives on the possible financial opportunities for solar PV investors in Cameroon derived from a simulation of the Cameroonian setting. It is a significant study for solar PV developers and investors in Cameroon where the feasibility of a large-scale solar PV project is assessed, analyzing the inherent risk associated with the investment. This is needed to erase the flawed perception that solar power investment in Cameroon is risky and expensive, one of the reasons why its deployment is slow.

1.2. Cameroon's solar energy potentials

The theoretical national average of electricity production potential from solar energy in Cameroon is estimated at 2327.5 TWh which is about 20 times the hydroelectric potential (19.7 TWh; Ministry of Energy and Water Resources—Cameroon, 2015; SIE-Cameroon, 2011). Cameroon's geographical position near the equator exposes the country to a significant amount of solar energy with mean irradiance of 4.5 kWh/m²/day southward and 5.74 kWh/m²/day north of the country. The mean insolation in the country is 4.9 kWh/m²/day (Muh et al., 2017). The solar potential in Cameroon can be exploited in two forms; solar thermal and solar photovoltaic. Solar PV and Solar thermal compete to a certain extent with utility-scale applications. Nevertheless, the two technologies vary in particular aspects and would probably coexist due to their potential to complement each other (International Renewable Energy Agency (IRENA), 2012). Despite this potential, the installed

Figure 1. Solar irradiance map of Cameroon (Solargis, 2022).



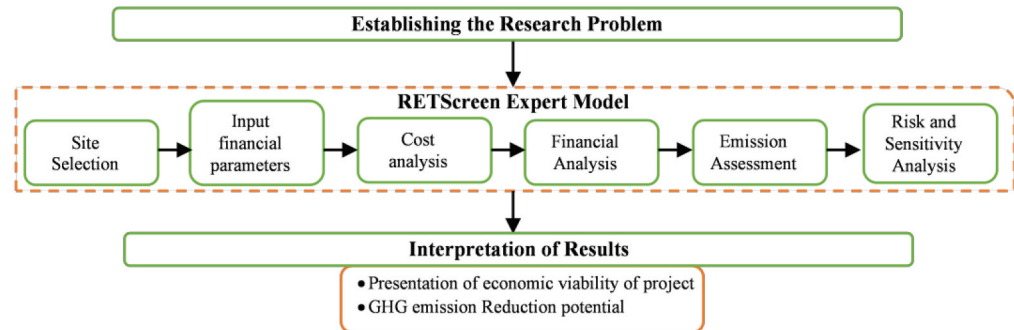
solar PV capacity is about 14.19 MW which are solely off-grid systems powering rural communities. Figure 1 shows the solar irradiance map of Cameroon.

2. Materials and methods

Climate data of Yaounde was obtained from NASA and financial inputs such as debt ratio, inflation rate, debt interest rate and discount rate were used in the analysis on the RETScreen Expert software (RETScreen, 2022). The analysis involved electricity generation assessment, life-cycle cost, GHG emissions analysis, GHG emissions reduction credit scheme, financial analysis, risk analysis and sensitivity analysis (Khalid & Junaidi, 2013; Mirzahosseini & Taheri, 2012; Sinha & Chandel, 2014). This analysis was based on an on-grid system where the feasibility of a grid-connected solar PV project was analyzed in Yaounde, Cameroon. The main indicators used in making decisions were the payback period, NPV, total investment cost, electricity exported to the grid and internal rate of return (IRR).

The financial analysis was based on assumptions gotten from available literature (International Renewable Energy Agency, 2018; International Renewable Energy Agency (IRENA), 2019b; International Renewable Energy Agency (IRENA), 2021). These values were obtained from IRENA reports with values extracted for the year 2020. An on-grid generation cost of 0.017 \$/kWh was used in the analysis (Energy Sector Management Assistance Program (ESMAP), 2014; Pappis, 2016). The evaluation of the project investment cost was conducted in the financial analysis so as to develop the most comprehensive financial model for the solar PV project. The GHG emission analysis used the Cameroon's electricity matrix by considering the fuels used for power generation to calculate the GHG emissions that can be evaded on substituting fossil fuels with renewables (RETScreen, 2022). The method applied in this study is similar to that used in (Mehmood et al.,

Figure 2. Methodology of the study.



2014; Mirzahosseini & Taheri, 2012; Rashwan et al., 2017; Rehman et al., 2017) with the only difference being the inclusion of the sensitivity/risk analysis and the GHG emission reduction credit scheme. For the few scholars who used the sensitivity/risk assessment such as (Bustos et al., 2016; Harder & Gibson, 2011), this study has included the GHG emission reduction credit scheme. Figure 2 shows the method adopted in the study.

2.1. The RETScreen model

Several models have been used in assessing the technical and financial viability of power projects. The RETScreen Expert tool offers a systematic framework for the estimation of power generation, financial viability, potentials of GHG emission reduction and project cost (RETScreen, 2022). The tool contains standard modules that can be used to assess the production of electricity, life-cycle cost of power projects and the GHG emission reduction potential of RE technologies. The approach used by this model is a top-down method with less input data requirements and offers a low-cost initial evaluation of RE projects. Unlike HOMER which uses hourly solar radiation data for one year (8760 data points), RETScreen models use monthly average solar radiation with only 12 data points (Tisdale et al., 2014). Nonetheless, comparative studies between RETScreen and models that use hourly data points as an alternative to monthly data has shown similar results, with a yearly variance of less than 5 % for anticipated energy generation (RETScreen, 2015). The RETScreen tool has a rich data base of integrated power plants, weather data as well as the ability to generate comparable results.

In RETScreen Expert, the PV system energy yield was assessed through the use of the annual average irradiation, humidity coefficients, azimuth, temperature and technical data such as system efficiency and losses. This energy estimate was further used to determine the project's GHG emission reduction potential and financial feasibility. In analyzing the GHG reduction potential, RETScreen Expert calculates the annual GHG emission reduction potential for the RE technology compared to a base case (local grid). Transmission and distribution (T&D) losses are equally accounted for in RETScreen. Figure 3 shows the benchmark analysis of the RETScreen simulation.

The viability of RE technologies can be determined by the initial capital costs. This cost varies globally as a result of different taxation structures, custom duties, or market changes. The size of the project equally has a major impact on the cost of investment because of scale effects. Generally, larger projects have a lower cost of investment per kW than smaller projects. The solar irradiation is essential for the analysis of the PV system. To accurately forecast the power generation from the solar PV system, RETScreen needs site-specific global solar irradiance data. Monthly average data was used in the analysis and they were assumed to be constant throughout the years for simplicity. The average monthly solar irradiation in Yaounde, Cameroon was 4.67 kWh/m²/day with a peak value of 5.49 kWh/m²/day in the month of February. A fixed mounting system was used in this study with an optimum tilt angle of 5 degrees. Figure 4 shows the schematic of the on-grid solar PV system.

Figure 3. Benchmark analysis of the solar PV project.

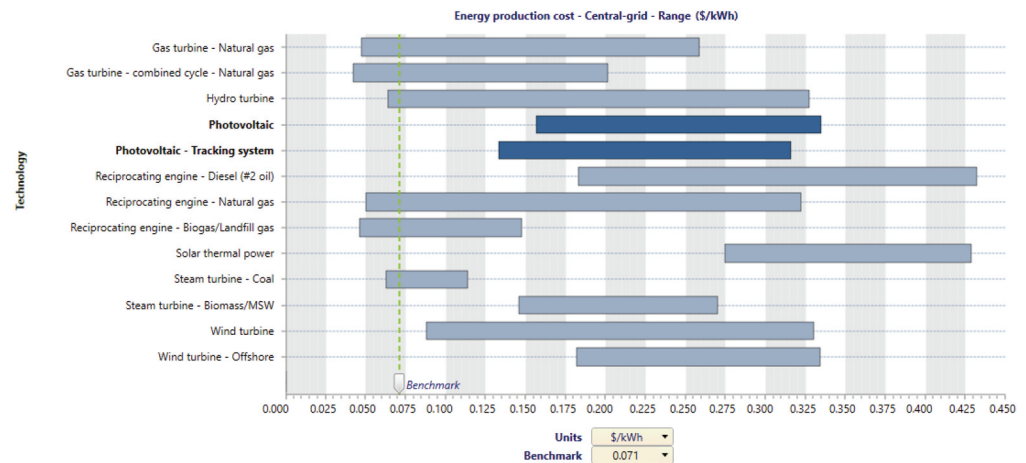
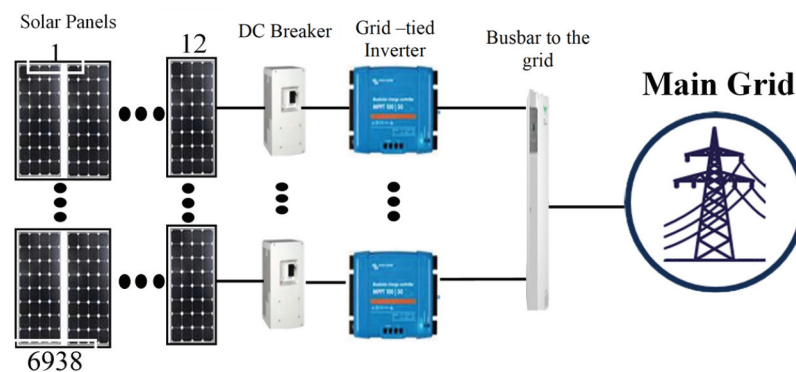


Figure 4. Schematic of the grid-connected PV system.



2.2. Site of solar PV project

The location Yaounde was selected so as to obtain the weather data of the solar PV system. The selected site is at latitude 3.9° N and longitude 11.5° E. The solar irradiation, humidity, average annual temperature and elevation are considered. The model used the meteorological data of the selected site to calculate the generated power injected into the grid and the system's capacity factor. Figure 5 shows a Google image of the location.

The carbon analysis was done by calculating the annual GHG emission reductions for the power project and comparing it to a base case. RETScreen has a data base of historical emission factors of grids in different countries obtained from the Intergovernmental Panel on Climate Change (IPCC; Intergovernmental Panel on Climate Change (IPCC), 2022a). The base case is assessed by considering the electricity mix by fuel type and baseline T&D losses. The T&D losses for Cameroon is 38% (Energy of Cameroon (ENEO), 2020) but a value of 24% was used in the analysis because the utility company is already putting measures to reduce the T&D losses. The emission factor for energy supply in Cameroon was assumed to be 0.263 tCO₂ /MWh or 0.263 kgCO₂ /kWh (Intergovernmental Panel on Climate Change (IPCC), 2022b). RETScreen conducts a comparison between the baseline GHG emission value and the proposed system (PV system in this case). The PV system is not considered to emit GHGs in this model. However, this model does not contain a complete life—cycle analysis of the PV system, as there will obviously be GHG emissions in the production process of PV accessories such as transportation of raw materials, resource extraction, and product fabrication. The feasibility of the grid connected solar PV was conducted for Yaounde with available satellite data from RETScreen's data base. Table 1 shows the site's characteristic.

Figure 5. Google image showing the location of Yaounde.

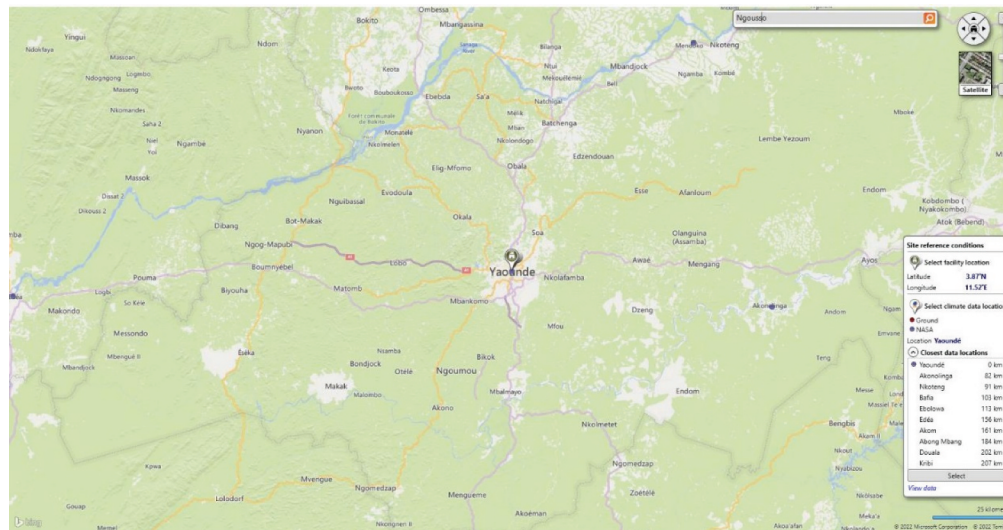


Table 1. Geographical data of the site

Parameter	Value
Latitude (°)	3.9
Longitude (°)	11.5
Climate zone	2A—Hot—Humid
Elevation (m)	622
Heating design temperature (°C)	18.4
Heating design temperature (°C)	27.9
Earth temperature amplitude (°C)	6.5

The heating design temperature is the least temperature of a particular area, measured at a frequency level of at least 1% throughout the year while cooling design temperature is the highest temperature of a particular area, measured for a frequency level of at least 1% throughout the year. The earth temperature amplitude is half of the difference between the highest and least earth temperature at the depth of measurement.

3. Results and discussion

The RETScreen Expert tool was used to evaluate the feasibility of implementing a 211.75 MW grid-connected solar PV system in Yaounde, Cameroon. The location's climatic data was obtained from RETScreen's database as shown in Table 2. Yaounde has an annual solar radiation of 4.69 kWh/m²/d where the month of July had the least average solar radiation and January has the highest solar radiation.

3.1. Electricity generation

The model was based on the feasibility section of the RETScreen Expert tool and the proposed solar PV capacity was estimated at 211.75 MW which is the power that will be injected into the Southern Interconnected Grid (SIG) of Cameroon. However, due to the intermittent nature of solar radiation, it is challenging to harness constant power from the solar PV system. An optimum tilt of 5 degrees was obtained through variation and monitoring of power production to maximize the generated power. This tilt was selected because it produced the largest solar irradiance as well as the highest electricity export to the grid for that location at a fixed panel orientation of 5 degree. The azimuth

Table 2. Average monthly solar PV electricity exported to the grid

Month	Daily solar radiation—horizontal (kWh/m ² /d)	Daily solar radiation—tilted (kWh/m ² /d)	Electricity export rate (USD/MWh)	Exported electricity to the grid (MWh)
January	5.43	5.60	120	30,705.445
February	5.49	5.59	120	27,612.007
March	5.20	5.22	120	28,605.911
April	4.97	5.92	120	26,121.531
May	4.65	4.55	120	25,065.398
June	4.26	4.15	120	22,315.829
July	4.00	3.92	120	21,865.969
August	3.98	3.93	120	21,953.848
September	4.27	4.26	120	22,969.308
October	4.14	4.18	120	23,285,740
November	4.56	4.67	120	25,023,389
December	4.12	5.29	120	29,143.816
Annual	4.67	4.69	120	304,668.191

considered in the study was 0 degree. Table 2 shows the monthly solar PV generated electricity that was exported to grid. In this analysis, the annual average solar radiation was 1.70 MWh/m².

From Table 2, it was observed that the annual energy exported to the grid was 304,668.191 MWh. The solar PV system has a total capacity of 211.75 MW which is made up of 770,000 solar panels of 275 Wp and a total grid-tie inverter capacity of 212 MW used in connecting the solar PV system to the grid. The Suntech mono-Si-STP2755-20/Wew solar module was used in the study. The characteristics of the solar PV system are presented in Table 3.

3.2. Financial analysis of solar PV system

Conducting financial calculations in order to assess project viability is a crucial part of project development as it guides investors during decision making. There are essential input parameters inserted in the model that led to outputs such as annual income, project cost, NPV, income summary, yearly cash flows and cumulative cash flow graphs. In order to conduct the financial analysis, the initial capital and annual cost assumptions for PV systems are required by the RETScreen tool. The capital cost and the annual fixed cost of solar PV used in the study were

Table 3. Properties of the inverter and the Solar PV system

Parameter	Specification
Slope	5°
Azimuth	0°
Power capacity	211.75 MW
Type of solar panel	mono-Si
Model	mono-Si-STP2755-20/Wew
Number of modules	770,000
Efficiency	16.9%
Nominal operating cell temperature	45°
Temperature coefficient	0.4 % /°C
Solar collector area	1,252,959 m ²

respectively 1378 \$/kW and 17.91 \$/kW/year (International Renewable Energy Agency, 2018; International Renewable Energy Agency (IRENA), 2019b; International Renewable Energy Agency (IRENA), 2021). An on-grid generation cost of 0.017 \$/kWh was used in the analysis (Energy Sector Management Assistance Program (ESMAP), 2014; Pappis, 2016). The RETScreen tool also required data for the the inflation rate, debt ratio, debt term, discount rate and the debt interest rate. These values differ based on the type of technology, plant capacity and location. Table 4 shows the financial parameters which have been considered in the study.

The data in Table 4 shows the solar PV has a project lifespan of 25 years with no grants awarded to the project. A debt ratio of 60:40 was used in the simulation, implying that equity amounts to 60% while the debt amounts to 40% of the initial capital cost. It was assumed that the debt will be repaid within a period of 15 years at an interest rate of 12% (Institut National de la Statistique (INS), 2016). In Cameroon, the tariff charged by Independent Power Producers (IPPs) for the sale of bulk electricity to the utility has been assumed to be \$ 0.12/kWh. After selecting the electricity export rate of \$0.12/kWh for the 25-year project life, the annual electricity price escalation rate was set to 0% owing to stable electricity prices in Cameroon for the last 11 years. An inflation rate of 3% was equally chosen for the financial analysis. As observed in Table 4, the total annual revenue obtained by the investor for exporting power to the grid was \$36,560,183 at an annual energy export of 304,668 MWh. There are possibilities that the projected annual electricity export could increase within the project life and hence, a projected annual electricity export rate of 2% was used in the analysis. It should be noted that this value does not consider the probable credit from the reduction of GHG emissions per year. This shows that solar energy could play a significant role in diversifying the Cameroonian power mix and generate huge positive impacts on the country's economy. Since the generated energy will directly be exported to the SIG, no fuel cost was added in the analysis. There are inadequate regulations in Cameroon that supports the deployment of renewables. However, government has established some economic incentives such as the tax exemption on the importation of solar panels and accessories (Cameroon National Assembly, 2011), as well as most recently giving a 10-year tax holiday (Ngalame, 2022) to any investor willing to invest in solar PV deployment in the country. Still, Cameroon lacks adequate regulations dedicated to solar PV systems and this is a huge barrier in the development of large-scale PV projects in the country. Table 5 shows the assessment of economic viability of the solar PV project.

The results in Table 5 show the economic viability of the solar PV project with a cost of energy (COE) of \$75.43/MWh or \$0.075/kWh which is equivalent to 48.75 FCFA (far less than the 82 FCFA tariff for commercial users in Cameroon). The benefit-cost ratio obtained in the simulation was 4.5

Table 4. Financial parameters in the model

Parameter	Value
Inflation rate (%)	3
Project lifetime (year)	25
Debt ratio (%)	40
Debt interest rate (%)	12
Debt term (year)	15
Debt payment (USD/year)	17,136,826
Total initial cost (USD)	291,791,500
Total annual cost (USD)	20,929,269
O&M cost (USD)	3,792,443
Electricity export rate (USD/MWh)	120
Electricity exported to grid (MWh)	304,668
Electricity export escalation rate (%)	2
Electricity export revenue (USD)	36,560,183

Table 5. Economic viability of the solar PV project

Item	Value
Pre-tax IRR—equity (%)	12.5
Pre-tax IRR—assets (%)	7.4
Simple payback (year)	8.9
Equity payback (year)	9.2
Net present value (USD)	619,909,244
Annual life cycle savings (USD/year)	24,796,370
Benefit-Cost (BC) ratio	4.5
Debt service charge	1.9
Energy production cost (USD/kWh)	0.075

which implies that the project is profitable since the ratio is greater than 1. The IRR and the break-even point of equity payback was 9.2 years while the simple payback was 8.9 years. It was seen that the PV system's simple payback was far less than the system's life expectancy. The PV system had an IRR-assets of 7.4%, which is below the debt interest rate of Cameroonian financial institutions. Therefore, the project will be attractive to investors. The feasibility analysis shows that Cameroon's pledge to increase the RE share by restructuring her electricity matrix can be realized through intensive investments in solar PV systems. This diversification would expand her economy through the reduction of fossil fuel-based thermal power plants and curtailing the country's overdependence on hydroelectricity. Harnessing solar energy will not only lessen fossil fuel usage for power generation but could potentially generate new openings in research, economic growth and employment in the Cameroonian society. Though current electricity regulations in Cameroon supports the expansion of RE projects, the lack of a specific laws on solar PV exploitation is a major obstacle in achieving a swift rise in solar PV installation in the country. This is obvious as most developing countries, like Cameroon, are financially constrained and cannot meet the financial implications (Mukisa, Zamora, Lie et al., 2021) of adopting practical policies that could advance clean energy. However, there is still hope for these countries through internationally sourced funds such as the green climate fund (GCF) and the clean development mechanism (CDM; Mukisa, Zamora, Tjing-Lie et al., 2021). These funds are acquired upon presentation of energy projects with the potentials to avoid GHG emissions. This is an indication that government policies on improving the deployment of clean energy in Cameroon should be backed by financial commitment. The cumulative and annual cash flows are presented in Figures 6 and 7 respectively.

3.3. Carbon analysis

For the carbon analysis, the GHG emission factor which is part of the RETScreen Expert data base was used in the assessment. During the carbon analysis, the RETScreen Expert tool calculates the amount of GHG emission reduction by means of using solar PV as a power generation source. The T&D losses used in the analysis was 24 % and this gave a GHG emission factor of 0.263 tCO₂ /MWh or 0.263 kgCO₂ /kWh. Figure 8 shows the results of the emission analysis.

The analysis showed that a 211.75 MW solar PV plant in Yaounde, Cameroon would lead to an annual GHG emission reduction of 61,004.5 tCO₂, which is the same as avoiding the utilization of 141,870.9 barrels of crude oil annually. The emission study of the proposed solar PV project reveals a GHG emission reduction equivalent to CO₂ emissions from 11,173 cars and light trucks, or 5,610.8 ha of CO₂ sequestration by forests. The details of the GHG emission reduction potentials are presented in Table 6.

The Food and Agricultural Organization (FAO) report on the Global Forest Resources Assessment (FRA) stated that the total world forest area reduced from 31.6 % to 30.6 % of the global land

Figure 6. Cumulative cash flow of the project.

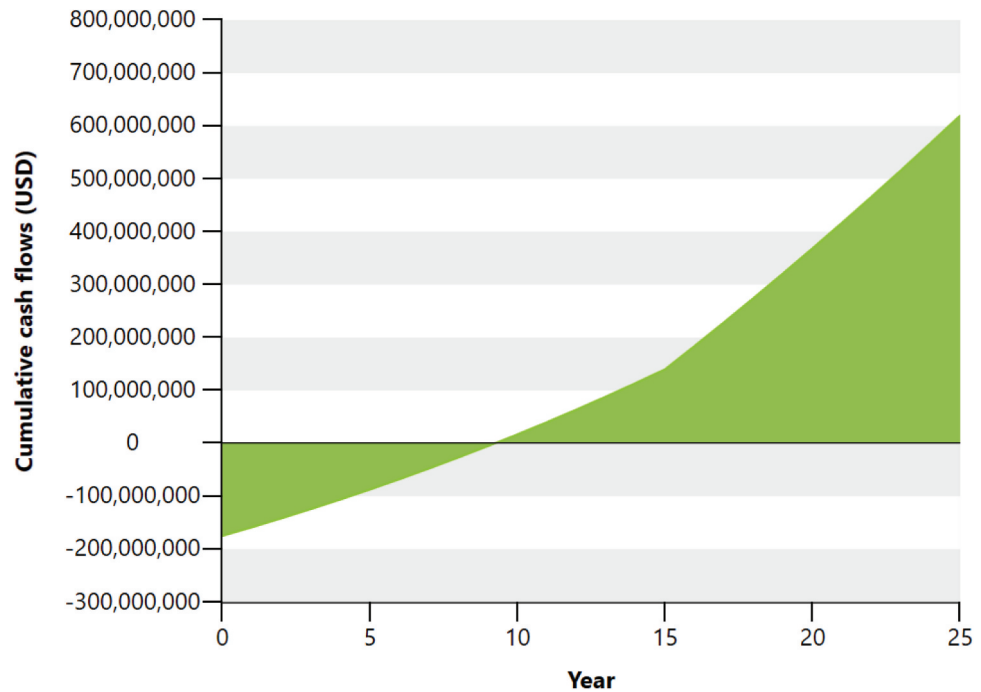
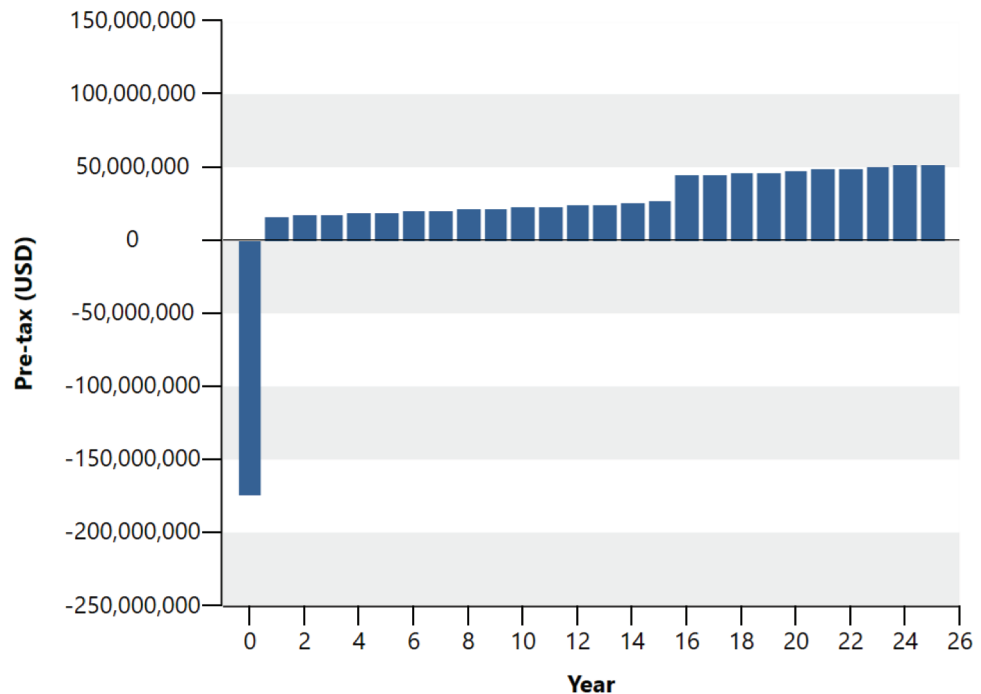


Figure 7. Annual cash flow of the project.



mass between 1990 and 2015 (Food and Agricultural Organisation (FAO), 2018). However, the development of clean energy technologies has reduced the rate of forest loss in the last few years. The deployment of this solar PV project advances Cameroon's quota of meeting the UN's sustainable development goals (SDGs) by 2030 through either directly or indirectly tackling the challenges of poverty, hunger and climate change. The project will indirectly mitigate these challenges through erecting resilient communities with reliable electricity, attaining inclusive growth and the sustainable management of the earth's natural resources. The results indicate that solar PV

Figure 8. The GHG emission reduction potential of solar PV project.

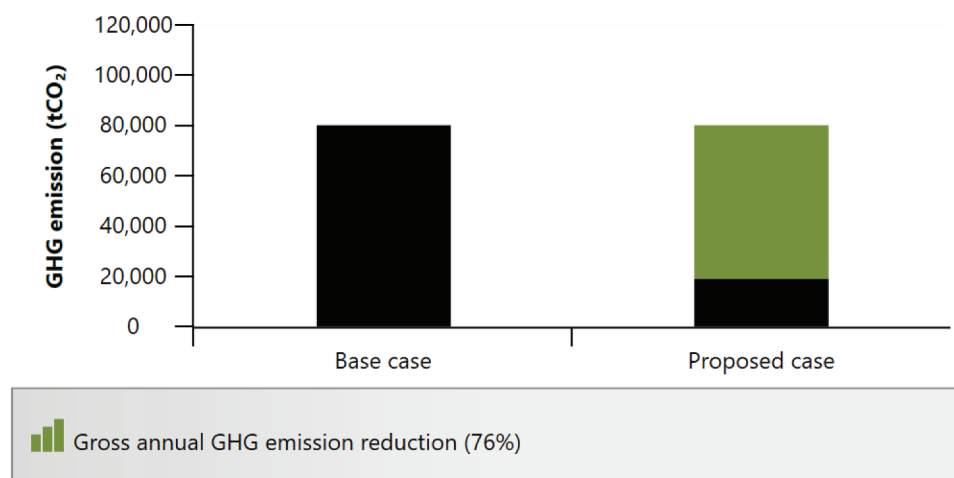


Table 6. Carbon analysis of the solar PV Project

Emission Reduction Pathway	Quantity
Avoided number of Cars and light trucks	11 173
Avoided use of Litres of gasoline	26 211 924.3
Avoided Barrels of crude oil	141 870.9
Number of people reducing energy use by 20%	61 004.5
Hectares of forest absorbing carbon	5610.8 Ha
Tons of waste recycled	21 036

deployment could significantly enhance Cameroon's climate commitments as well as her ambitions to diversify electricity generation and reduce the over dependence on hydropower which is increasingly posing future problems due to droughts. Hence, there is an urgent need to upscale investment in the solar energy sector as the diversification in the electricity generation matrix would improve economic development through reducing the frequent load shedding and making available more power for industrial expansion. This will greatly increase productivity as power shortages are affecting a lot of local industries in the country. The effect of power shortage was evident in the recently organized African Nations Cup 2022 in Cameroon, where many industries were asked to reduce productivity as electricity was mostly serving the residential customers and football stadia. Investing in solar PV will also reduce the importation of fossil fuels used for power generation in the country and hence, assist in redirecting the finances to other developmental projects. The study supports Cameroon's ambition of becoming a major power exporter in the Central African Power Pool (CAPP) as well as her preparation for future energy challenges.

3.4. GHG Emission reduction credit

This section presents the credit of GHG reduction through the installation of solar PV system. It signifies the credit of avoiding 1 ton of CO₂ emitted by a typical power plant in Cameroon. A carbon credit of 5 \$/tCO₂ was used in the calculation of the total credit. It should be noted that there are currently no carbon credit schemes in Cameroon. However, this analysis was to demonstrate possible revenue streams that could be added to clean technologies in Cameroon so as to encourage investors. This section is equally attempting to initiate policy discussions by stakeholders toward adopting this scheme. Table 7 shows the project's GHG emission reduction revenue.

The amount of 5 \$/tCO₂ was used to calculate the possible revenue from GHG emission trading in energy projects within the country. The credit value for GHG reduction was \$5/tCO₂ over a period of

Table 7. Emission reduction credit

Parameter	value
GHG reduction credit rate (USD/tCO ₂)	5
GHG reduction credit duration (year)	15
GHG reduction credit escalation rate (%)	1
Net annual GHG emission reduction (tCO ₂)	61,004.5
GHG reduction revenue (USD)	305,022

15 year at an annual escalation rate of 1%. The RETScreen Expert tool calculates the revenue generated from GHG emission reduction of the solar PV project. The profits that will be obtained from the 211.75 MW solar PV project was \$ 305 022.

3.5. Sensitivity analysis

The sensitivity analysis is the assessment of how changes in some input variables could affect the feasibility of the project. It helps to minimize the degree of uncertainty in a project through varying inputs while monitoring how they affect the financial variables of the project.

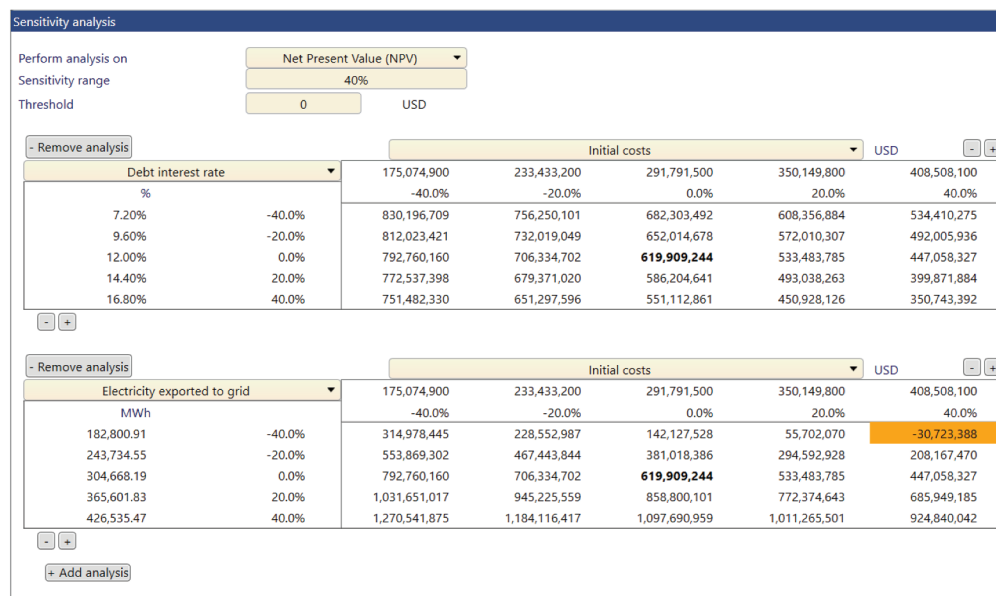
A sensitivity analysis was also conducted on the RETScreen model in order to determine the economic inputs needed to render the solar PV plant investment-friendly. This involved keeping some inputs constant and continuously changing one input variable while monitoring the effects on the output variables. Essential economic factors such as payback period, net present value (NPV) and internal rates of return were greatly influenced by solar irradiance of the location, capital cost, electricity export escalation rate, electricity export rate and RE production income. A positive NPV signifies that the project is economically viable. The analysis of the sensitivity of the project's NPV was conducted through varying the project initial cost versus the debt interest rate at a sensitivity range of $\pm 40\%$ in scenario 1. Scenario 2 involved varying the project initial cost versus the grid electricity export using a similar sensitivity range. Before the sensitivity analysis, the project initial cost was \$291,791,500 and after the $\pm 40\%$, the initial cost became \$408,508,100 and \$175, 074,900 respectively representing -40 and $+40$. Besides, the project's actual debt interest rate was 12%. However, after the application of the $\pm 40\%$ sensitivity range, it became 16.8% and 7.2% respectively. While keeping other parameters constant, the model recalculated the NPV for all the combinations of initial cost and debt interest rate. The RETScreen Expert tool presented the NPV values which are below the zero-threshold using the orange colour as seen in Figure 9. From Figure 9, it is seen that all the recalculated NPV values for scenario 1 are positive. This shows the solar PV project is financially viable for the $\pm 40\%$ sensitivity range when assessing the initial cost versus debt interest rate. As one observes the change in NPV values across rows and columns in Figure 9, there is a great difference in NPV values from one row to another than within the same column. This shows that, for this project, the NPV is influenced more by the initial cost than the debt interest rate.

For scenario 2, as the simulation was conducted at a 40% rise in the initial cost and a decrease of 40% in the grid electricity export, the project will not be financially viable because the NPV becomes $-\$30,723,388$ (section in orange colour in Figure 9). However, as both the grid electricity exported and project initial cost rises by 40%, the solar PV project will be economically feasible because the NPV becomes \$924,840,042 which is far greater than zero. Once more, this implies that the project's NPV has a higher sensitivity to the project's initial cost than the grid electricity export. Figure 9 shows the results of the sensitivity analysis.

3.6. Risk analysis

Similar to sensitivity analysis, the assessment of project risk is essential before the start of any project in order to identify factors that could affect the project's economic life. Risk analysis differs from sensitivity analysis in that, sensitivity analysis uses 2 parameters to conclude while risk analysis uses

Figure 9. Sensitivity analysis of the solar PV project.



all the parameters at a particular range to make conclusions. In this section, the energy production cost was chosen to be used as the financial indicator for the analysis. A range of $\pm 25\%$ was considered for all the factors. The RETScreen Expert tool performed a Monte Carlo simulation with 5000 number of combinations. This recalculates the energy production cost and generates the impact chart as well as the distribution graph as presented in Figures 10 and 11. From Figure 10, changes in the cost of energy production was caused by variations of the other system factors. It is observed that the project initial cost and the grid electricity export greatly impacted the solar PV project. The project initial cost is directly proportional to the cost of energy production while the electricity exported to the grid is inversely proportional to the cost of energy production. In Figure 11, the risk level is set at 10% without including the anticipated lower 5% and the upper 5% of the cost of energy production to

Figure 10. Solar PV Impact graph of risk analysis.

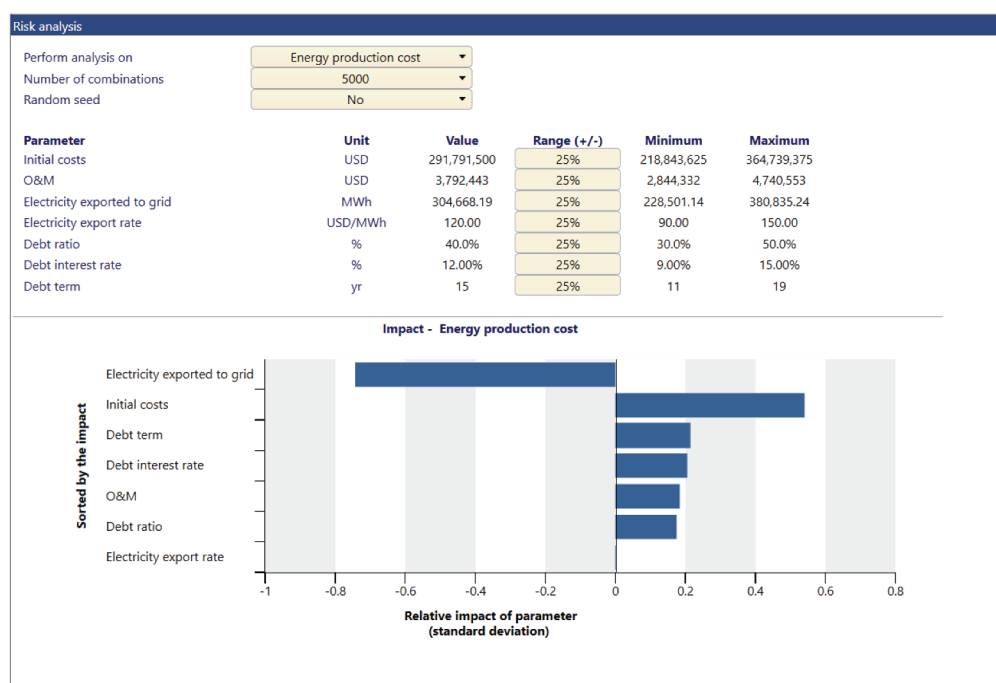
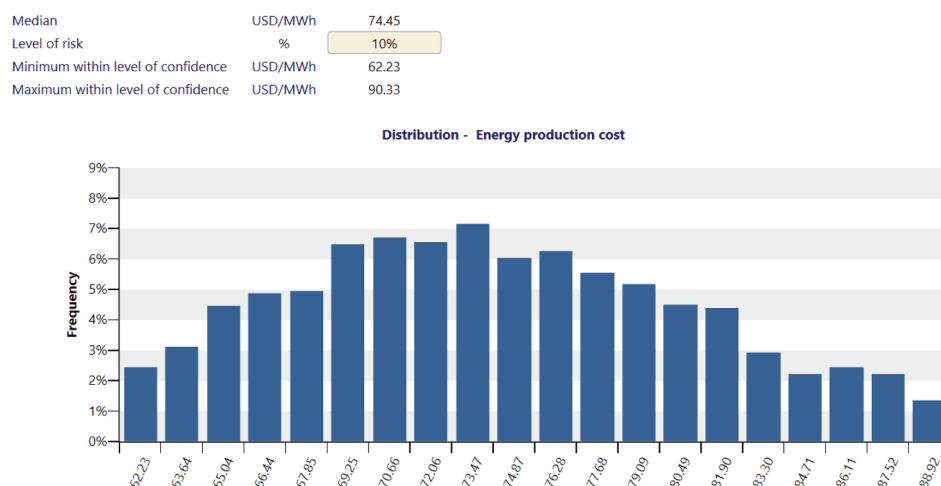


Figure 11. Distribution chart of the risk analysis of the solar PV.



surpass the P90 value (90% confidence level). The results showed that the P90 cost of energy production was 90.33 \$ /MWh which was closer to the actual cost of energy production of 75.43 \$/MWh obtained before the risk analysis, rendering the solar PV project bankable and realistic.

4. Conclusions

A detailed feasibility analysis of a 211.75 MW grid-connected solar PV was conducted in order to assess the project's viability in Cameroon through examining the risk, technical, sensitivity, financial and the environmental impact on Cameroon. The aim was to advance the need for government to pursue a solar power road map that is appealing for both national and international solar energy investors. The RETScreen Expert model was used and the results obtained includes the PV system energy yield, NPV, IRR, payback period, GHG emission reduction potential and the GHG emission mitigation credit for the PV project. The software equally computed the energy exported to the grid, system initial cost, the hectares of forest absorbing carbon, the project life cycle, GHG emission revenue as well as the project's financial wellbeing. The economic analysis had thoroughly presented the anticipated revenue from the electricity export to the grid and the component of the risk that could affect the solar PV project. The main findings include;

- The total annual revenue obtained from the solar PV project for exporting power to the grid was \$36,560,183 at an annual energy export of 304,668 MWh.
- The solar PV project's cost of energy (COE) was \$75.43/MWh or \$0.075/kWh which is equivalent to 48.75 FCFA/kWh. This is cheaper than the electricity price of 84 FCFA/kWh for commercial users (Electricity Sector Regulatory Agency (AESEL), 2012) in Cameroon.
- The benefit-cost ratio obtained in the simulation was 4.5 which implies that the project is profitable since the ratio is greater than 1.
- The break-even point of equity payback was 9.2 years while the simple payback period was 8.9 years. This shows that the PV system's simple payback is far less than the system's life expectancy of 25 years.
- The PV system had an IRR-assets of 7.4%, which is below the debt interest rate of Cameroonian financial institutions. Therefore, the project will be attractive to investors.
- The solar PV system had a gross annual GHG emission reduction of 61 004.5 tCO₂ corresponding to 141,870.9 barrels of crude oil not used throughout the project life time.
- The income envisaged, if Cameroon adopts the GHG emission reduction credit scheme of \$5/tCO₂ over a period of 15 year, from the 211.75 MW solar PV project will be \$ 305 022.
- The project risk and sensitivity analysis showed that a reduction in the project initial cost has a significant effect on the NPV and the cost of energy production. Also, a rise in the grid electricity export reduces the cost of energy production when the confidence level is set to 90%.

Due to the impressive viability of the solar PV project, the project is highly recommended for adoption and installation. Generally, it is clear from the analysis that achieving Cameroon's ambitious targets of generating 25% of renewable electricity in the power mix is possible. The financial analysis was conducted without any grants or subsidies from the government. There are prospects that some economic factors could change in future such as the falling price of PV modules and the possibilities of rising electricity tariff in Cameroon. These aspects are going to further make the solar PV project attractive to investors. The falling prices of solar PV technologies as well as the increasing cost of fossil-fuel based power generators will further render the use of renewables more competitive even in the absence of subsidies. It is worthy to note that economic changes such as rising inflation will negatively impact the project's viability. This is considered as a limitation to the findings presented in this study. In a long run, the cost of power generation in Cameroon will be reduced as well as the country's GHG emissions since RE systems will occupy most of the power generation mix. It is recommended that the Yaounde City Council should not rely on the central government to pursue the implementation of this project. The investment indicators for this project are quite bankable that the Yaounde City Council, with the recent decentralization of municipalities, could source partnership agreement with the Rural Electrification Agency in lobbying solar energy investors to set up this project which could be used as an additional source of income for the council. This would not only generate profits but will help stabilize the grid as well as mitigate the effect of global warming through avoided GHGs emissions. Therefore, it can be concluded that the grid-connected solar PV offers several benefits such as; electricity cost saving, improved grid reliability, and reduced environmental pollution.

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